Linear Simultaneous Equations using Elimination

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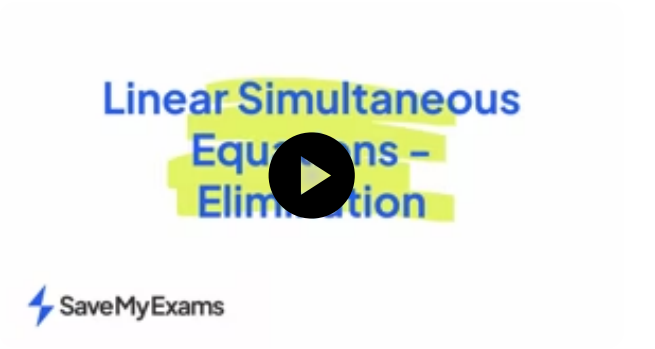

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Help me understand

Linear simultaneous equations using elimination

What are linear simultaneous equations?

- When you have more than one equation in more than one unknown, then you are dealing with **simultaneous equations**
- An equation is **linear** if none of the unknowns in it is raised to a power other than one
- Solving a **pair of simultaneous equations** means finding **pairs of values** that make **both equations true** at the same time
- A linear equation in two unknowns will produce a straight line if you graph it... linear = line
- A pair of simultaneous equations will produce lines that will cross each other (if there is a solution!)

How do I use elimination to solve linear simultaneous equations?

Step 1: Multiply one (or both) of the equations by a constant (or constants) to get the numbers in front of **one** of the unknowns to match

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

Step 3: Solve the new equation from Step 2 to find the value of one of the unknowns

Step 4: Substitute the value from Step 3 into one of the original equations, and solve to find the value of the other unknown

Step 5: Check your solution by substituting the values for the two unknowns into the original equation you **didn't** use in Step 4

Examiner Tips and Tricks

- Don't skip the checking step (it only takes a few seconds) – there are many places to go wrong when solving simultaneous equations!
- Mishandling minus signs is probably the single biggest cause of student error in simultaneous equations questions

Worked Example

Solve the simultaneous equations:

$$-2x + 4y = 5$$
$$4x - 5y = -7$$

$$-2x + 4y = 5 \quad (1)$$
$$4x - 5y = -7 \quad (2)$$

LABEL THE EQUATIONS

STEP 1: [2 × (1)] $-4x + 8y = 10$
[(2)] $4x - 5y = -7$

WE ONLY HAVE TO MULTIPLY ONE OF THE EQUATIONS HERE

STEP 2:
$$\begin{array}{r} -4x + 8y = 10 \\ +\ (4x - 5y = -7) \\ \hline 3y = 3 \end{array}$$

THE SIGNS IN FRONT OF THE 4's ARE DIFFERENT, SO ADD

STEP 3: $y = 3 \div 3 = 1$

STEP 4: $4x - 5(1) = -7$
$4x = -2$
$x = -0.5$

USING EQUATION (2)

STEP 5: $-2(-0.5) + 4(1) = 5$
$1 + 4 = 5$
$5 = 5$ √

USING EQUATION (1)

$x = -0.5 \quad y = 1$

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